

```
import os

# Install unrar if it's not already installed
if not os.path.exists('/usr/bin/unrar'):
    !apt-get update && apt-get install -y unrar

# Extract the rar file
!unrar x /content/rag_project.rar /content/

# Change to the extracted directory
%cd /content/rag_project
```

UNRAR 6.11 beta 1 freeware Copyright (c) 1993-2022 Alexander Roshal

Extracting from /content/rag_project.rar

```
Creating      /content/rag_project                OK
Extracting    /content/rag_project/chunker.py      OK
Extracting    /content/rag_project/config.py       OK
Extracting    /content/rag_project/data_crawler.py OK
Extracting    /content/rag_project/evaluator.py    OK
Extracting    /content/rag_project/main.py         OK
Extracting    /content/rag_project/rag_pipeline.py OK
Extracting    /content/rag_project/README.md       OK
Extracting    /content/rag_project/requirements.txt OK
Extracting    /content/rag_project/vector_store.py OK
Creating      /content/rag_project/data            OK
All OK
/content/rag_project
```

```
!pip install -q chromadb sentence-transformers openai tiktoken rank-bm25 matplotlib
```

验证安装

```
import chromadb
import sentence_transformers
from openai import OpenAI
print("✅ 依赖安装完成")
```

```

_____ 52.0/52
_____ 23.3/23.3
_____ 278.2/278
_____ 4.6/4.6 M
_____ 18.7/18.7
_____ 72.5/72.5
_____ 178.9/178
_____ 61.9/61.9
_____ 203.7/203
_____ 71.6/71.6
_____ 61.7/61.7
_____ 60.6/60.6
```

ERROR: pip's dependency resolver does not currently take into account all the packages that a
google-adk 2.3.0 requires opentelemetry-api<=1.42.1,>=1.39, but you have opentelemetry-api 1.
google-adk 2.3.0 requires opentelemetry-sdk<=1.42.1,>=1.39, but you have opentelemetry-sdk 1.

✅ 依赖安装完成

```
!python main.py process --max_tokens 512
```

```
06:09:49 [INFO] =====
06:09:49 [INFO] 处理数据: 分块 + 建索引
06:09:49 [INFO] =====
06:09:49 [INFO] 已加载 7 篇文章
06:09:49 [INFO] 加载 7 篇文章
06:09:50 [INFO] 分块完成: 7 篇 → 79 块
06:09:50 [INFO] 分块完成: 79 块
06:09:59 [INFO] NumExpr defaulting to 2 threads.
06:10:02 [INFO] TensorFlow version 2.20.0 available.
06:10:02 [INFO] JAX version 0.7.2 available.
06:10:07 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to ena
06:10:07 [WARNING] Warning: You are sending unauthenticated requests to the HF Hub. Please se
06:10:07 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] Loading SentenceTransformer model from BAAI/bge-small-zh-v1.5.
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:08 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
Loading weights: 100% 71/71 [00:00<00:00, 4583.59it/s]
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:09 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:10 [INFO] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-zh-v1.5/tr
06:10:10 [INFO] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-zh-v1.5/tr
06:10:10 [INFO] HTTP Request: HEAD https://huggingface.co/BAAI/bge-small-zh-v1.5/resolve/main
06:10:10 [INFO] HTTP Request: HEAD https://huggingface.co/api/resolve-cache/models/BAAI/bge-s
06:10:10 [INFO] HTTP Request: GET https://huggingface.co/api/models/BAAI/bge-small-zh-v1.5 "H
06:10:10 [INFO] Embedding 模型: BAAI/bge-small-zh-v1.5
Batches: 100% 3/3 [00:23<00:00, 7.76s/it]
06:10:34 [INFO] 向量数据库: 添加 79 个块 (总计 79)
06:10:34 [INFO] 索引完成: 79 个块
06:10:34 [INFO] 分块参数: max_tokens=512, overlap=64
```

```
%cd /content/rag_project
!python main.py pipeline --max_articles 30 --max_tokens 512 --n_questions 20 --n_ev
```

06:02:32 [INFO] X 内容为空或过短
06:02:35 [INFO] [2/30] 深度学习
06:02:35 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:35 [INFO] X 内容为空或过短
06:02:38 [INFO] [3/30] GPT-4o
06:02:38 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:38 [INFO] X 内容为空或过短
06:02:39 [INFO] [4/30] 随机森林
06:02:39 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:39 [INFO] X 内容为空或过短
06:02:42 [INFO] [5/30] 循环神经网络
06:02:42 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:42 [INFO] X 内容为空或过短
06:02:44 [INFO] [6/30] GPT-5
06:02:44 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:44 [INFO] X 内容为空或过短
06:02:46 [INFO] [7/30] 序贯博弈
06:02:46 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:46 [INFO] X 内容为空或过短
06:02:48 [INFO] [8/30] C4.5算法
06:02:48 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:48 [INFO] X 内容为空或过短
06:02:50 [INFO] [9/30] GPT-4
06:02:50 [INFO] ✓ 1099 字符
06:02:53 [INFO] [10/30] 注意力經濟
06:02:54 [INFO] ✓ 6160 字符
06:02:55 [INFO] [11/30] Seq2Seq模型
06:02:55 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:55 [INFO] X 内容为空或过短
06:02:57 [INFO] [12/30] 决策树剪枝
06:02:57 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:02:57 [INFO] X 内容为空或过短
06:02:59 [INFO] [13/30] Mamba2
06:03:00 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:03:00 [INFO] X 内容为空或过短
06:03:02 [INFO] [14/30] PDF軟體列表
06:03:02 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:03:02 [INFO] X 内容为空或过短
06:03:04 [INFO] [15/30] 图神经网络
06:03:04 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:03:04 [INFO] X 内容为空或过短
06:03:05 [INFO] [16/30] 残差神经网络
06:03:05 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:03:05 [INFO] X 内容为空或过短
06:03:07 [INFO] [17/30] 基于内容的图像检索
06:03:07 [INFO] ✓ 2221 字符
06:03:10 [INFO] [18/30] 聚类分析
06:03:10 [INFO] ✓ 11961 字符
06:03:12 [INFO] [19/30] 注意力机制
06:03:12 [WARNING] 请求失败: 403 Client Error: Too many requests. Please respect our robot
06:03:12 [INFO] X 内容为空或过短

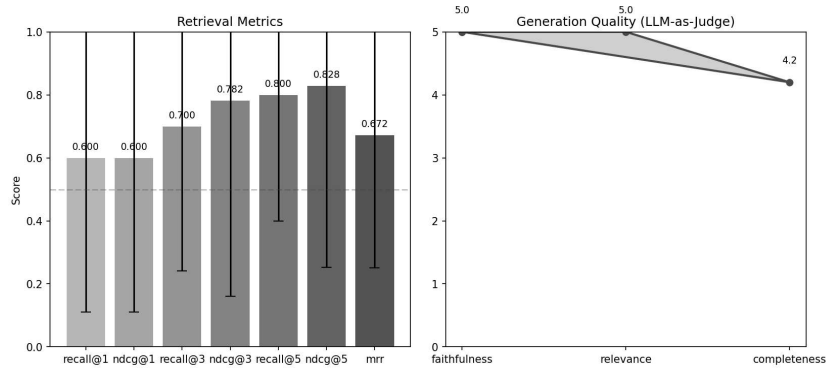
```
!python main.py eval --n_questions 20 --n_eval 10
```

```
recall@5: 0.8000 (std: 0.4000)
ndcg@5: 0.8281 (std: 0.5763)
mrr: 0.6725 (std: 0.4217)
06:14:07 [INFO]
[3/3] 评估生成质量 (LLM-as-Judge)...
06:14:07 [INFO] RAG Pipeline 初始化: model=deepseek-chat
Batches: 100% 1/1 [00:00<00:00, 32.55it/s]
06:14:07 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:09 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 40.89it/s]
06:14:10 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:11 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 32.71it/s]
06:14:13 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:14 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 35.78it/s]
06:14:15 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:17 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 36.03it/s]
06:14:18 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:22 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:22 [INFO] 生成评估: 5/10
Batches: 100% 1/1 [00:00<00:00, 31.81it/s]
06:14:23 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:25 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 38.29it/s]
06:14:26 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:28 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 29.02it/s]
06:14:29 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:30 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 45.37it/s]
06:14:32 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:34 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
Batches: 100% 1/1 [00:00<00:00, 35.58it/s]
06:14:35 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:36 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 "
06:14:37 [INFO] 生成评估: 10/10
06:14:38 [INFO] 生成评估: 忠实度=5.00, 相关性=5.00

--- 生成质量 ---
faithfulness: 5.00 (std: 0.00)
relevance: 5.00 (std: 0.00)
completeness: 4.20 (std: 0.87)
06:14:39 [INFO] 评估图表已保存: /content/rag_project/output/eval_results.png
Figure(1600x500)
06:14:39 [INFO] 评估完成! 结果已保存到 /content/rag_project/output
```

```
%cd /content/rag_project
from IPython.display import Image, display
display(Image(filename="output/eval_results.png"))
```

/content/rag_project



RAG Evaluation Summary

recall@1: 0.6000 $\pm$ 0.4899
ndcg@1: 0.6000 $\pm$ 0.4899
recall@3: 0.7000 $\pm$ 0.4583
ndcg@3: 0.7815 $\pm$ 0.6207
recall@5: 0.8000 $\pm$ 0.4000
ndcg@5: 0.8281 $\pm$ 0.5763
mrr: 0.6725 $\pm$ 0.4217

faithfulness: 5.00 $\pm$ 0.00
relevance: 5.00 $\pm$ 0.00
completeness: 4.20 $\pm$ 0.87

```
%cd /content/rag_project
```

```
# 实验1: 小分块
```

```
!python main.py eval --max_tokens 256 --n_questions 15 --n_eval 8
```

```
# 实验2: 大分块
```

```
!python main.py eval --max_tokens 1024 --n_questions 15 --n_eval 8
```

```
06:18:50 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
06:18:52 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
Batches: 100% 1/1 [00:00<00:00, 35.01it/s]
06:18:53 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
06:18:54 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
Batches: 100% 1/1 [00:00<00:00, 34.25it/s]
06:18:56 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
06:18:57 [INFO] HTTP Request: POST https://api.deepseek.com/chat/completions "HTTP/1.1 200 OK"
06:18:58 [INFO] 生成评估: 忠实度=5.00, 相关性=5.00
```

--- 生成质量 ---

faithfulness: 5.00 (std: 0.00)

relevance: 5.00 (std: 0.00)

completeness: 5.00 (std: 0.00)

```
06:18:59 [INFO] 评估图表已保存: /content/rag_project/output/eval_results.png
Figure(1600x500)
```

```
06:18:59 [INFO] 评估完成! 结果已保存到 /content/rag_project/output
```